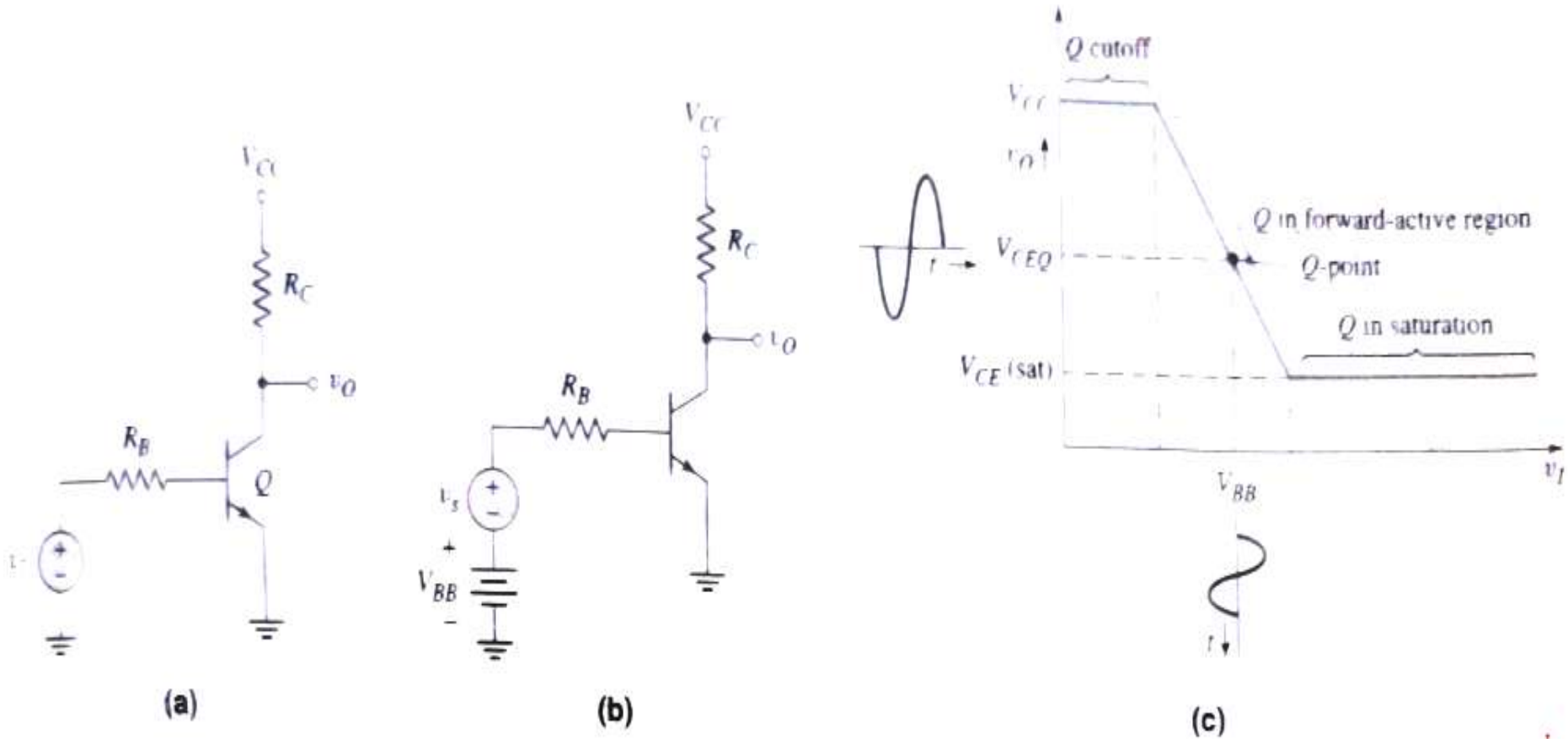
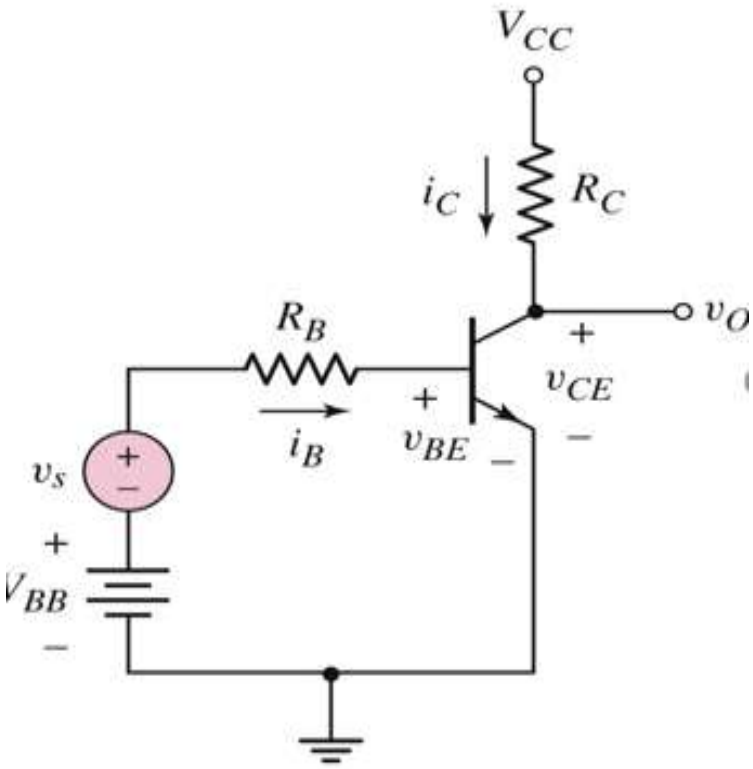
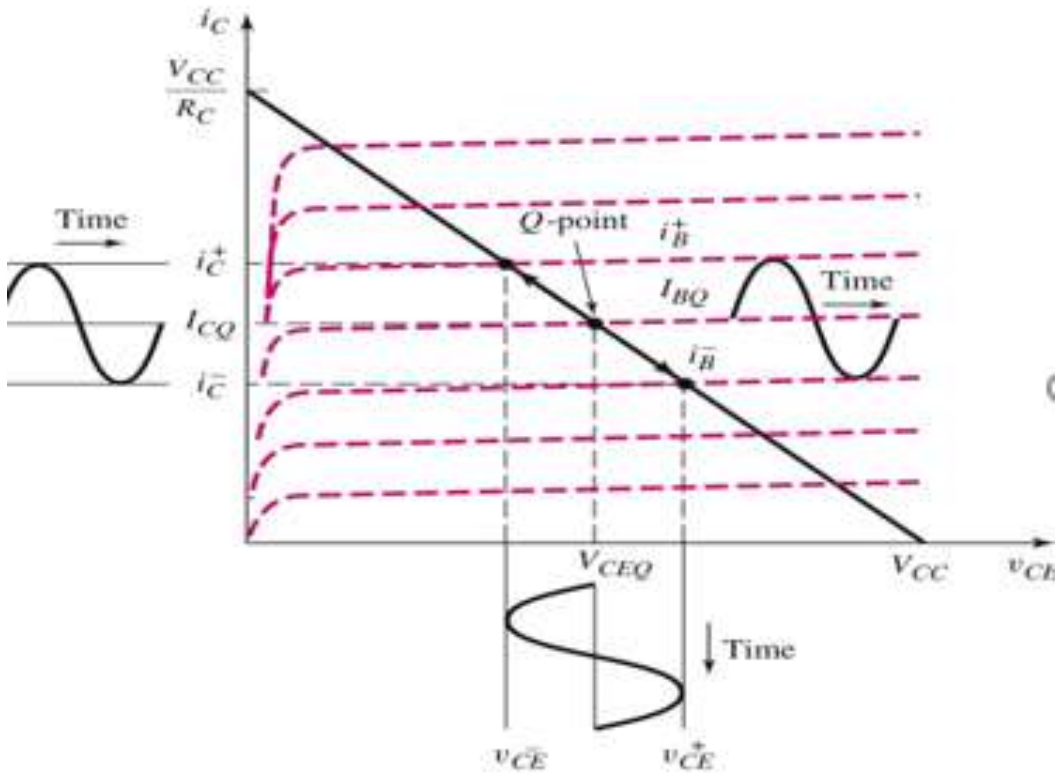


a. Bipolar transistor inverter circuit
b. Inverter circuit showing both dc bias and ac signal source
c. Transistor inverter voltage transfer characteristics showing desired Q-point

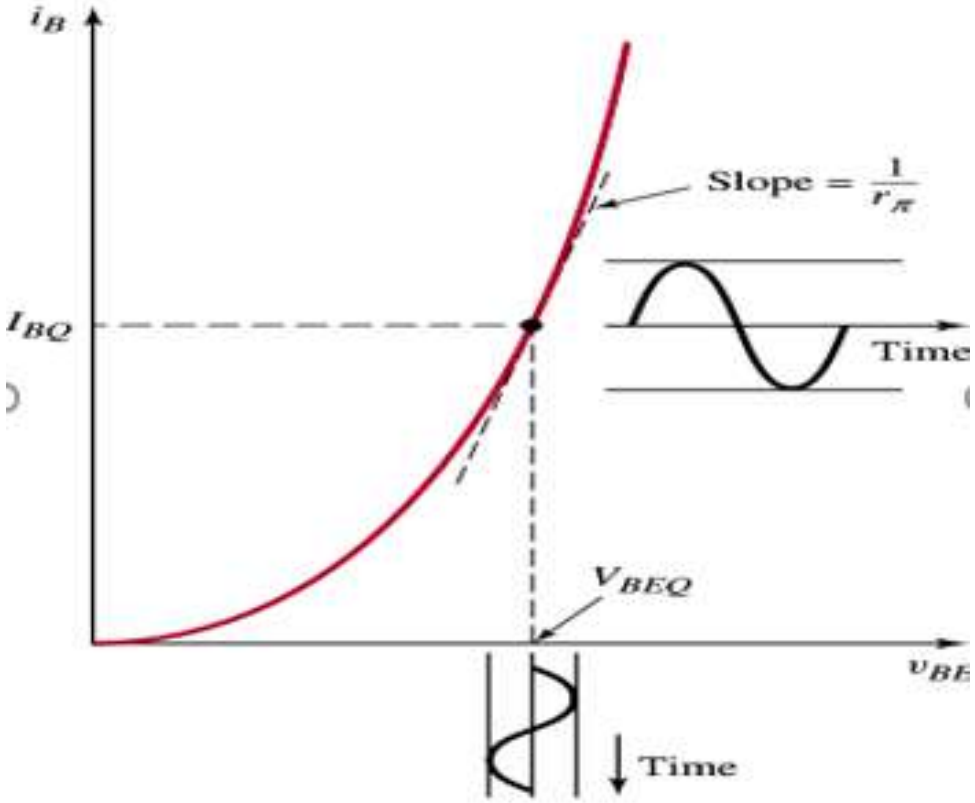
Small signal analysis of BJT Amplifier

Common Emitter with Time-Varying Input



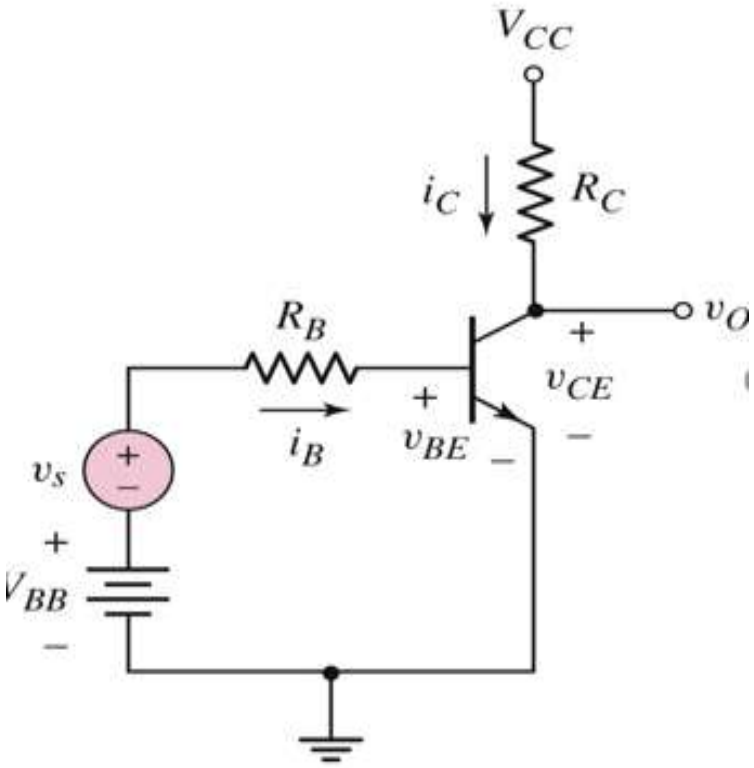


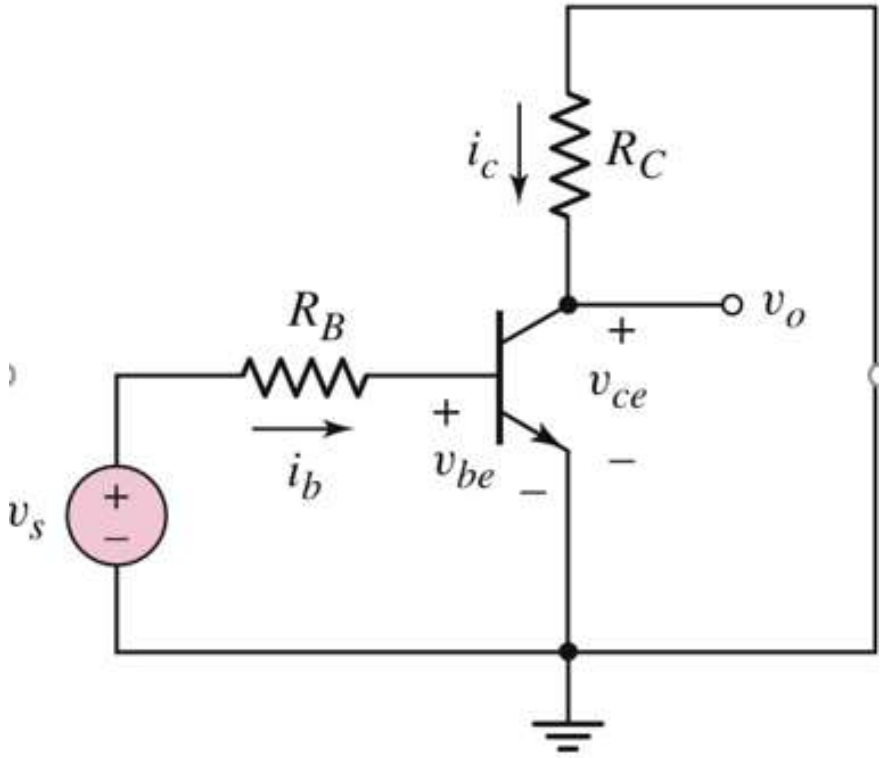
I_B Versus V_{BE} Characteristic



$$i_B \cong I_{BQ}(1+\frac{v_{be}}{V_T}) = I_B + i_b$$

ac Equivalent Circuit for Common Emitter





ac Equivalent Circuit for Common Emitter.
DC voltage sources have been set equal to zero

Small-Signal Hybrid π Model for npn BJT

The **Hybrid-π Model** is an equivalent circuit model used to analyze the **small-signal (AC)** behavior of a BJT (Bipolar Junction Transistor) operating in its active region.

The hybrid- **π** model replaces the non-linear physical BJT in your circuit diagram with simple linear circuit elements (a resistor and a controlled current source) so you can apply standard circuit rules like Kirchhoff's Laws.

Why create Model?

BJTs (Bipolar Junction Transistors) and MOSFETs are fundamentally non-linear.
If you look at a transistor's actual physical behavior, the relationship between input voltage and output current is exponential

$$I_C = I_S \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$

Trying to calculate circuit voltages and currents across multiple connected components using exponential equations directly is extremely difficult and quickly becomes mathematically intractable for complex circuits.

Why Model?

1. Linearization (Small-Signal Approximation)

When a transistor is used as an amplifier, it operates around a fixed DC point (the **operating point** or **Q-point**), with tiny AC signals riding on top of it.

On a large exponential curve, a tiny segment looks almost like a **straight line**.

Modeling replaces the complex exponential curve with a **linear approximation** at that specific Q-point.

Linear circuits allow you to apply simple mathematical tools like Ohm's Law, Kirchhoff's Voltage/Current Laws (KVL/KCL), Superposition, and Thevenin/Norton equivalents.

Why Model?

2. Separation of DC and AC Analysis

Modeling allows us to use the **Principle of Superposition** to break a complex circuit analysis into two manageable steps:

DC Analysis (Biasing): You turn off all AC sources (treat capacitors as open circuits) to find the DC bias current and voltage. This sets the parameters of your model (like transconductance $\mathbf{g_m}$.

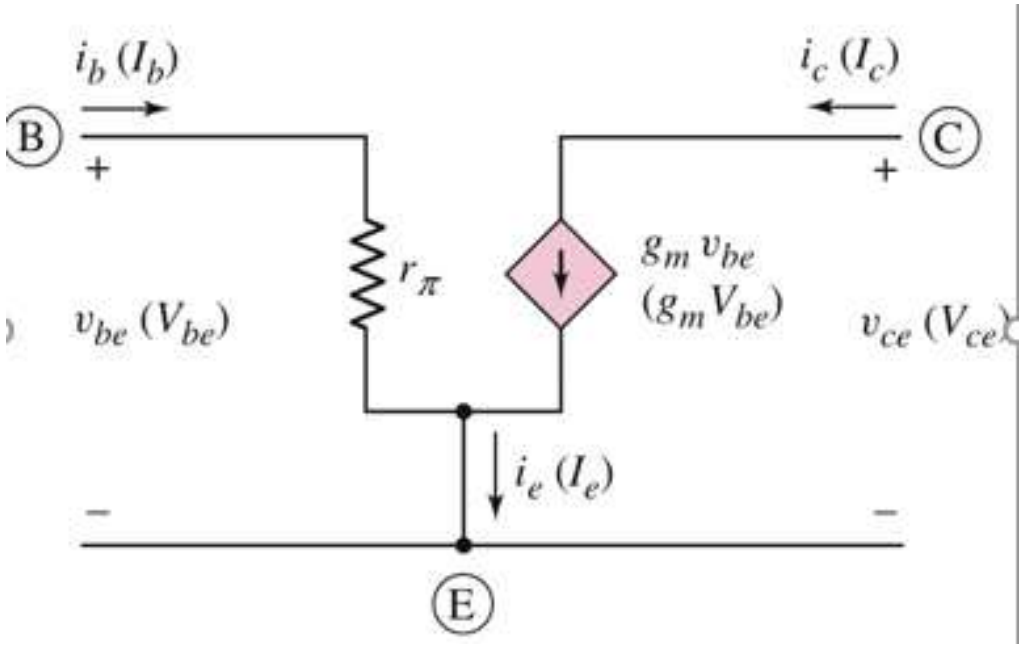
AC Analysis (Signal Processing): You replace the physical transistor with its equivalent model (using parameters calculated from the DC step), set DC supplies to ground, and easily compute gain, input impedance, and output impedance.

Small-Signal Hybrid π Model for npn BJT

$$g_m = \frac{I_{CQ}}{V_T}$$

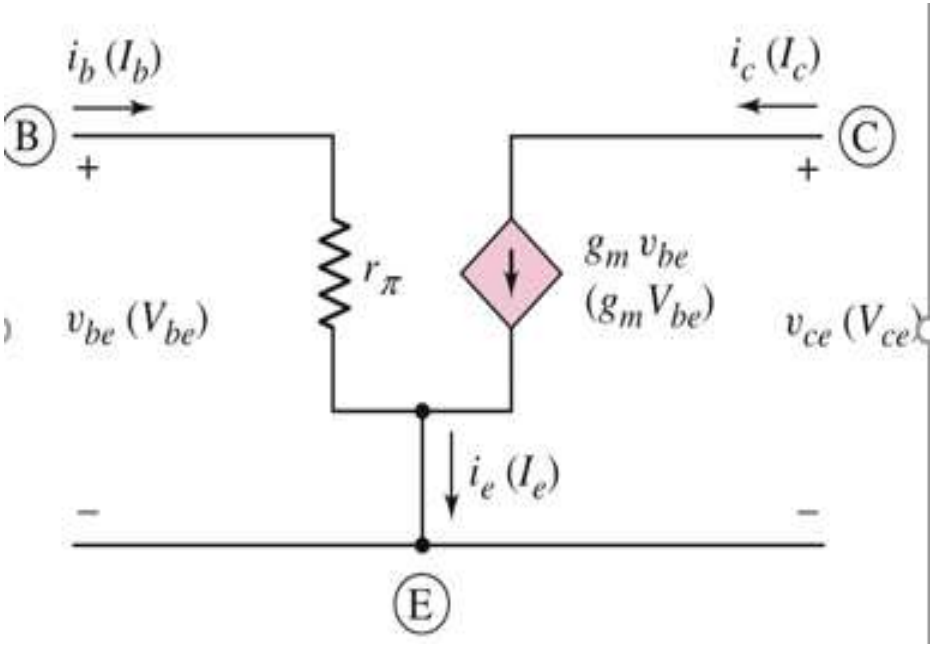
$$r_\pi = \frac{\beta V_T}{I_{CQ}}$$

$$g_m r_\pi = \beta$$



Small-Signal Hybrid π Model for npn BJT

Small-Signal Hybrid π Model for npn BJT

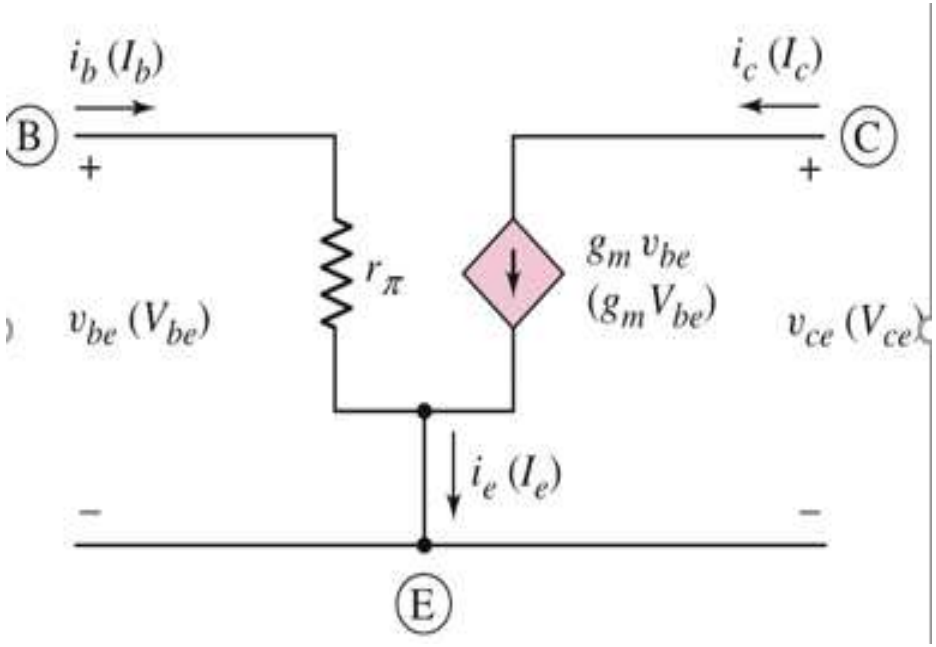


Small-Signal Hybrid π Model for npn BJT

The Input Side (Base-Emitter Terminal)

- v_{be}**(Base-Emitter AC Voltage):** The small-signal AC voltage applied across the Base and Emitter terminals.
- $\mathbf{r_\pi}$ **(Input Resistance / Dynamic Resistance):** The AC resistance looking into the base of the transistor. It models the forward-biased PN junction of the Base-Emitter.

Small-Signal Hybrid π Model for npn BJT

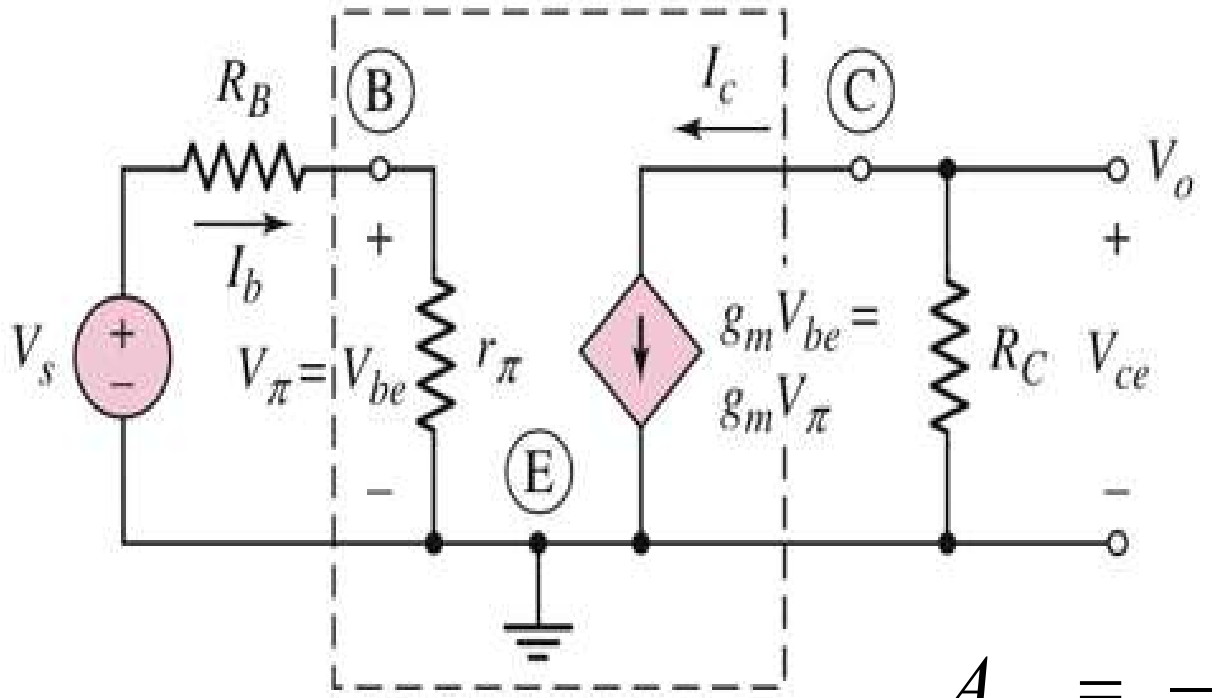


Small-Signal Hybrid π Model for npn BJT

2 The Output Side (Collector-Emitter Terminal)

$g_m\ v_{be}$: (Voltage-Controlled Current Source): Represents the amplification process. A small variation in the input voltage v_{be} controls the output current flowing through the collector terminal.

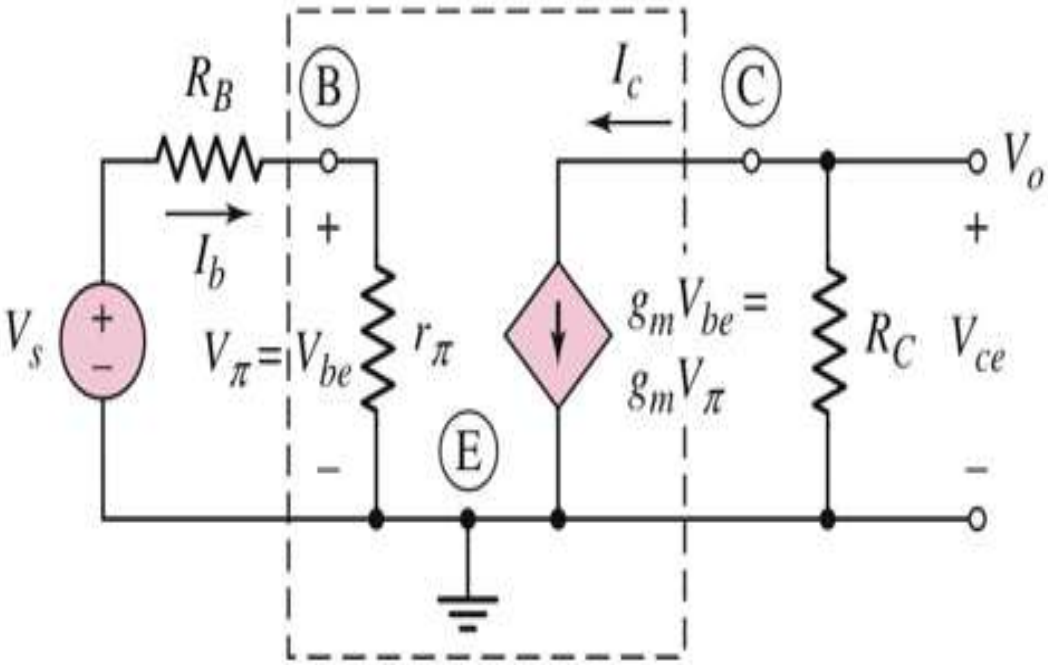
Small-Signal Equivalent Circuit for npn Common Emitter circuit



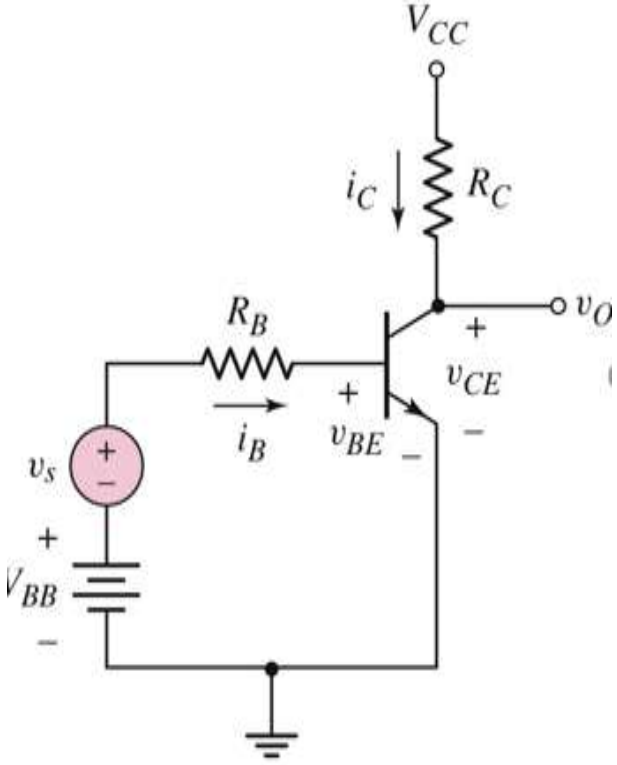
$$A_v = -(g_m R_C)\left(\frac{r_\pi}{r_\pi + R_B}\right)$$

Small Signal Voltage Gain

- Av = Vo/ Vs
- Vo = V_{CE} = - (g_m V_π)R_C
- $V_\pi = \left(\frac{r_\pi}{r_\pi + RB} \right) V_S$
- $V_S = V_\pi / \left(\frac{r_\pi}{r_\pi + R} \right)$
- Av = Vo/ Vs = - g_m R_C $\left(\frac{r_\pi}{r_\pi + RB} \right)$



- Calculate small signal voltage gain of circuit.
- $\beta = 100$, $V_{cc} = 12$ V, $V_{BE} = 0.7$V, $R_c = 6$ KΩ, $R_B = 50$KΩ and $V_{BB} = 1.2$V



DC Analysis:

$I_{BQ} = (V_{BB} - V_{BE}) / R_B = (1.2 - 0.7) / 50 = 10\ \mu A$

$I_{CQ} = \beta I_{BQ} = 100 \cdot 10\ \mu A = 1$ mA

$V_{CEQ} = Vcc - I_{CQ} Rc = 6$ V

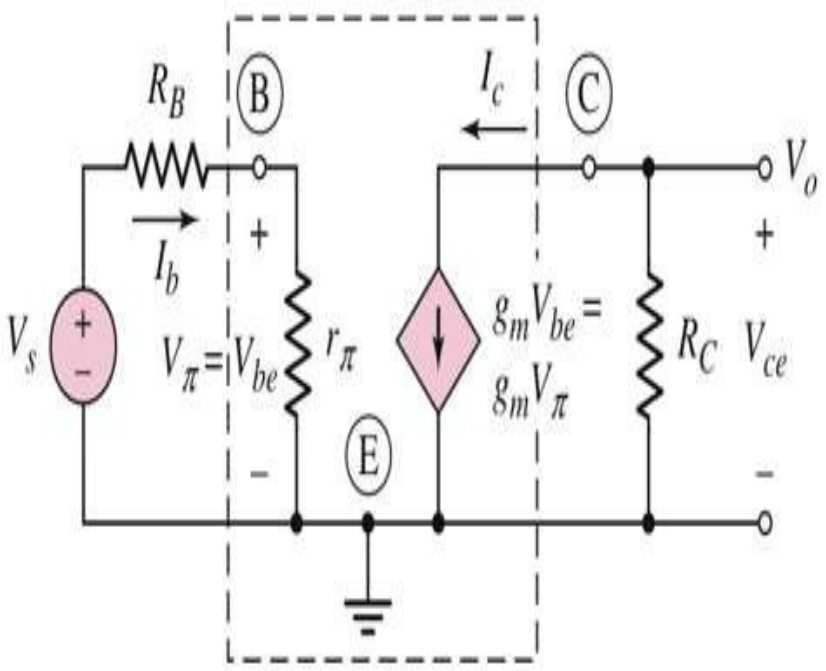
AC Analysis: $r_\pi = \dfrac{\beta V_T}{I_{CQ}}$

$= (100)(0.026)/1 = 2.6\text{K}\ \Omega$

$g_m = \dfrac{I_{CQ}}{V_T} = 1/0.026 = 38.5$ mA /V

$A_v = -(g_m R_C)\left(\dfrac{r_\pi}{r_\pi + R_B}\right) = -11.4$

Example with specific input



Consider , Vs = 0.25 sin ωt V

$$i_b = \frac{Vs}{r_\pi + RB}$$

$$= \frac{0.25 \sin \omega t}{50+2.6}$$

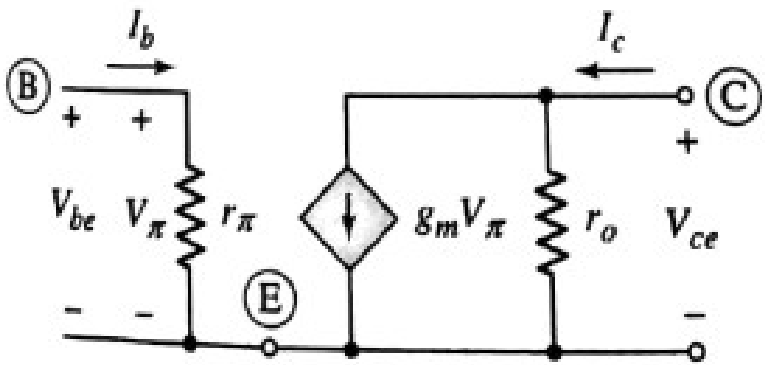
$$= 4.75 \sin \omega t \ \mu A$$

$$i_c = \beta i_b = (100)*(4.75 \sin \omega t)$$

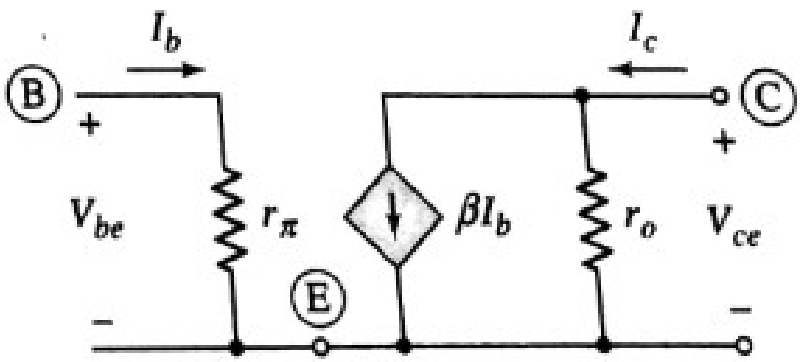
$$= 0.475 \sin \omega t \ mA$$

$$v_{ce} = - i_c Rc = -(0.475 \sin \omega t)6$$

$$= -2.85 \sin \omega t \ V$$



(a)



(b)

$$g_m = \frac{I_{CQ}}{V_T}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}}$$

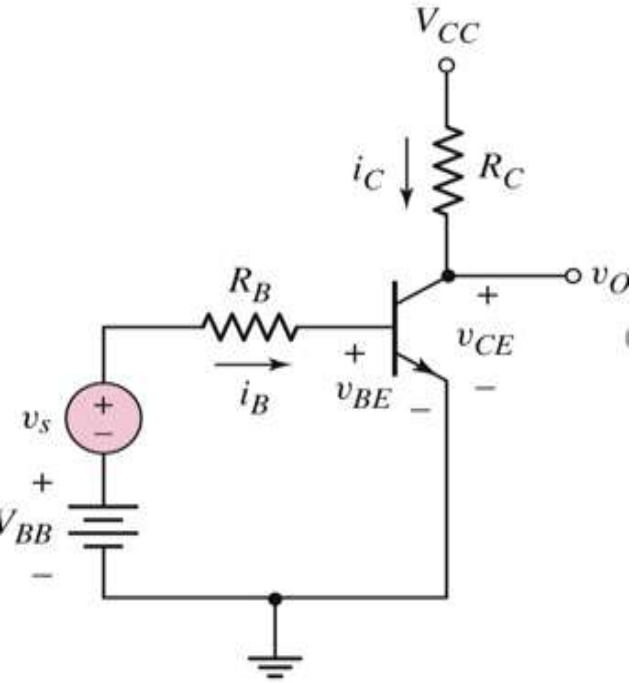
$$g_m r_\pi = \beta$$

$$g_m \frac{V_\pi}{I_b} = \beta$$

$$g_m V_\pi = \beta I_b$$

EX 6.1:Calculate small signal hybrid **π parameters,** voltage gain of circuit, Q pt

β =100, Vcc =5 V, $V_{BE(ON)}$ = 0.7V, Rc =15 KΩ, R_B=650KΩ and V_{BB} =2V



DC Analysis:

$I_{BQ} = (V_{BB} - V_{BE}) / R_B = (2 - 0.7) / 650 =$ **0.002 m A**

$I_{CQ} = \beta I_{BQ} =$ 100 (0.002) μ A **= 0.2 mA**

$V_{CEQ} = Vcc - I_{CQ} Rc = 2$ V

AC Analysis:

$$r_\pi = \frac{\beta V_T}{I_{CQ}}$$

$=$ (100) (0.026)/0.2 =**13 K Ω**

$$g_m = \frac{I_{CQ}}{V_T} = 0.2/\ 0.026 =$$ **7.7 mA /V**

$$A_v = -(g_m R_C)\left(\frac{r_\pi}{r_\pi + R_B}\right)$$

$=$ **-2.26**

Bipolar AC analysis

- As we are dealing with linear amplifier, superposition applies. It means that we can perform dc and ac analysis separately.
  - 1. Analyse circuit with only dc sources present. Calculate Q point. Transistor must be biased in forward active region in order to produce a linear amplifier

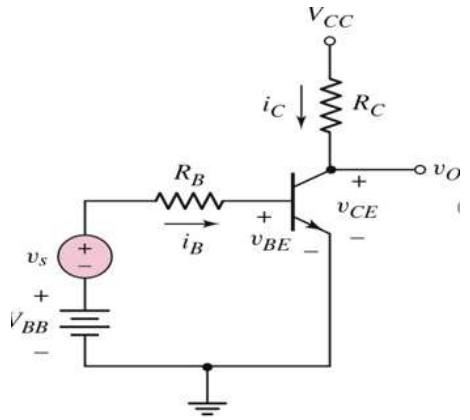
Element	*I–V* relationship	DC model	AC model
Resistor	$I_R = \frac{V}{R}$	R	R
Capacitor	$I_C = sCV$	Open	C
Inductor	$I_L = \frac{V}{sL}$	Short	L
Diode	$I_D = I_S(e^{v_D/V_T} - 1)$	$+V_\gamma - r_f$	$r_d = V_T/I_D$
Independent voltage source	V_S = constant	$+V_S-$	Short
Independent current source	I_S = constant	I_S	Open

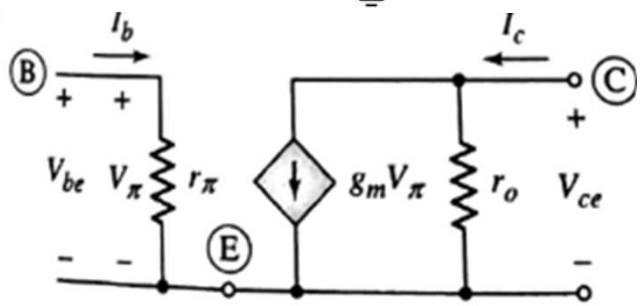
2. Replace each element in circuit, with its small signal model. Transistor by hybrid π model.

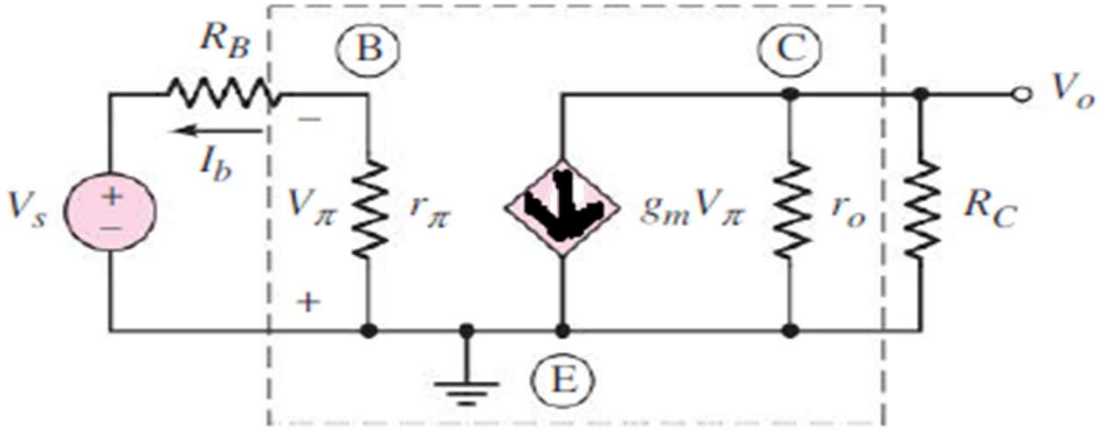
3. Analyse the small signal equivalent circuit , setting dc source components to zero.

CE amplifier Circuit with r_o

- Av = Vo/ Vs
- Vo = V_{CE}
- Vo = V_{CE} = - (g_m V_π) (R_C \\ r_o)
- $V_\pi = \left(\frac{r_\pi}{r_\pi + RB} \right) V_S$
- $V_S = V_\pi / \left(\frac{r_\pi}{r_\pi + RB} \right)$
- Vo/ Vs = - g_m (R_C \\ r_o) $\left(\frac{r_\pi}{r_\pi + RB} \right)$

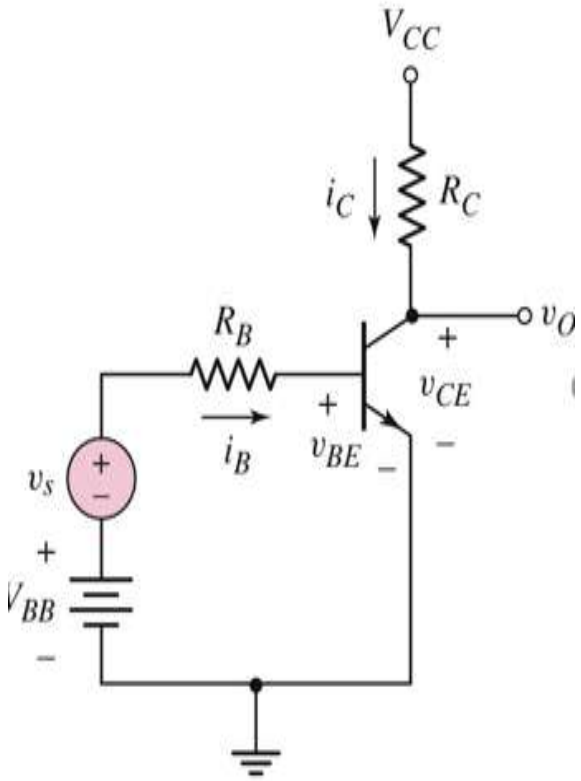






Calculate small signal voltage gain of circuit, including effect of r_o.

β =100, Vcc =12 V, V_{BE} = 0.7V, Rc =6 KΩ, R_B=50KΩ and V_{BB} =1.2V V_A = 50 V



$$r_o = \frac{V_A}{I_{CQ}} = 50 \text{ K}$$

DC Analysis:

$I_{BQ} = (V_{BB} - V_{BE}) / R_B = (1.2 - 0.7) / 50 = 10\ \mu A$

$I_{CQ} = \beta I_{BQ} = 100 . 10\ \mu A = 1$ mA

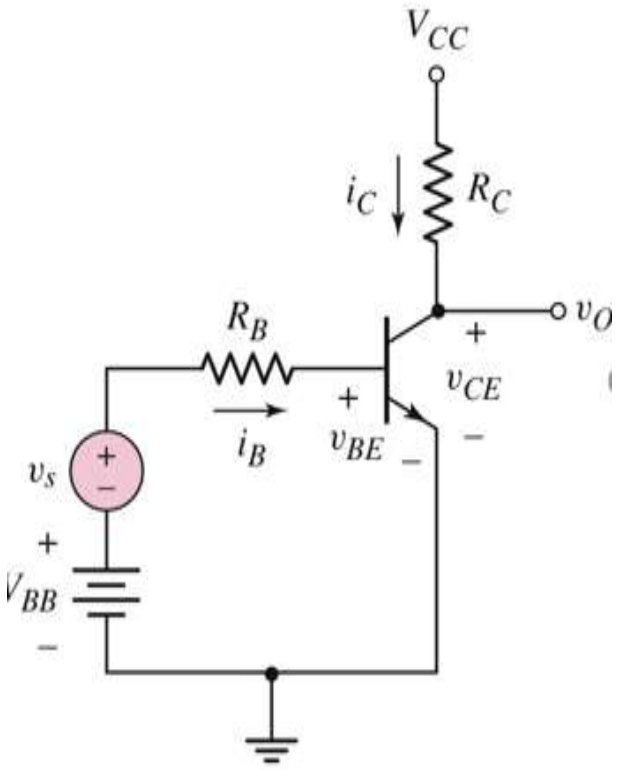
$V_{CEQ} = Vcc - I_{CQ} Rc = 6$ V

$$Av = Vo/Vs = - g_m (Rc \,//\, r_o) \left(\frac{r_\pi}{r_\pi + R_B} \right) = -10.2$$

r_o reduces magnitude of Av.

If ro >>Rc, then effect of ro is negligible.

- Calculate small signal voltage gain of circuit without ro , with ro
- β =100, Vcc =12 V, V_{BE} = 0.7V, Rc =6 KΩ, R_B=50KΩ and V_{BB} =1.2V V_A =50V



DC Analysis:

$I_{BQ} = (V_{BB} - V_{BE}) / R_B = (1.2 - 0.7) / 50 = 10\ \mu A$

$I_{CQ} = \beta I_{BQ} = 100 \cdot 10\ \mu A = 1\ mA$

$V_{CEQ} = Vcc - I_{CQ} Rc = 6\ V$

AC Analysis: $r_\pi = \frac{\beta V_T}{I_{CQ}} = (100)(0.026)/1 = 2.6K\ \Omega$

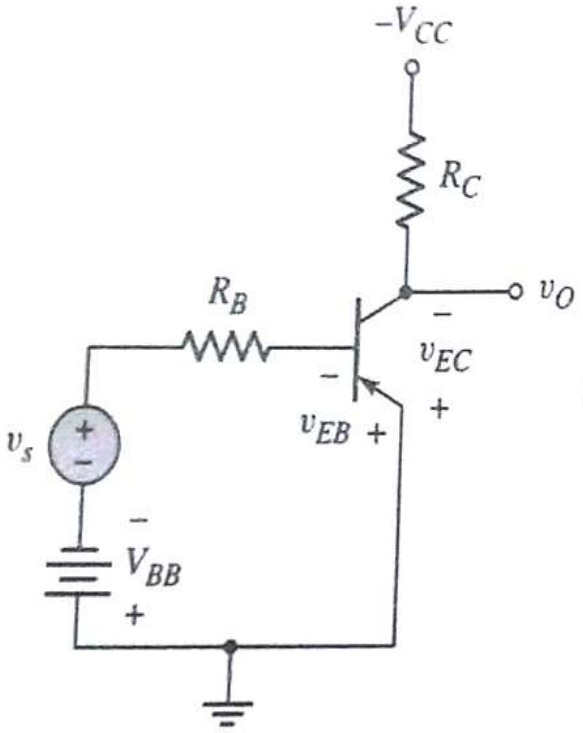
$g_m = \frac{I_{CQ}}{V_T} = 1/0.026 = 38.5\ mA/V$

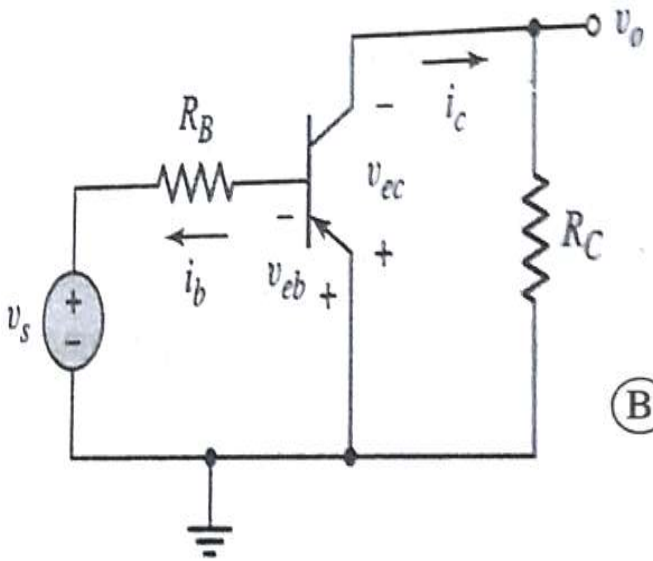
$$A_v = -(g_m R_C)\left(\frac{r_\pi}{r_\pi + R_B}\right) = \mathbf{-11.4}$$

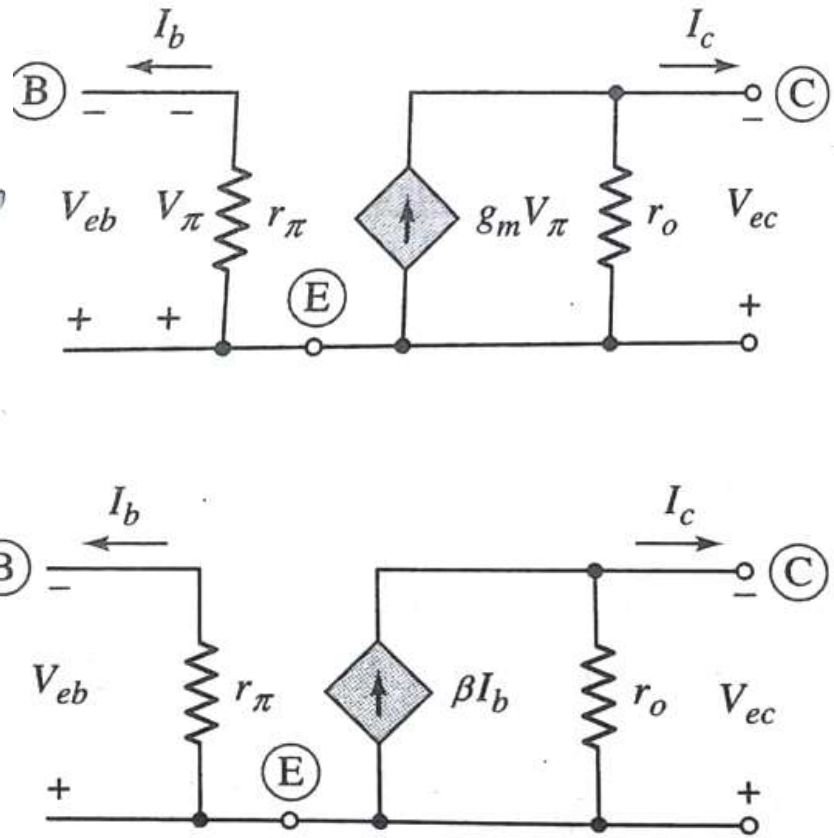
$$r_o = \frac{V_A}{I_{CQ}} \qquad r_o = \frac{V_A}{I_{CQ}} = \frac{50}{1m} = 50K\Omega$$

$$Av = -g_m (R_C \,\backslash\backslash\, r_o)\left(\frac{r_\pi}{r_\pi + RB}\right) = \mathbf{-10.2}$$

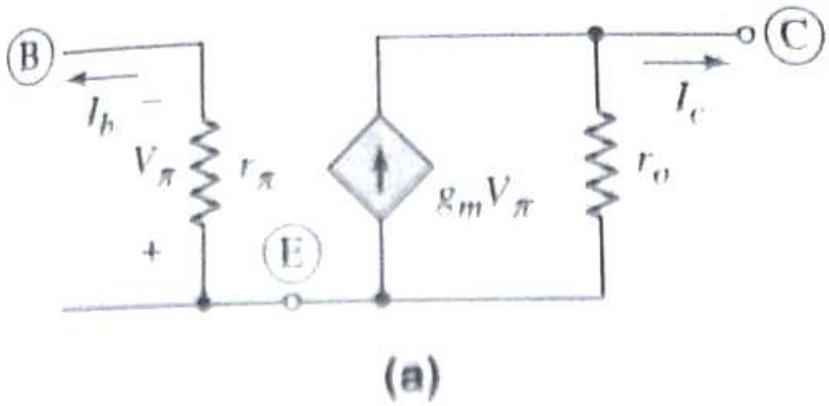
A common emitter circuit with pnp transistor



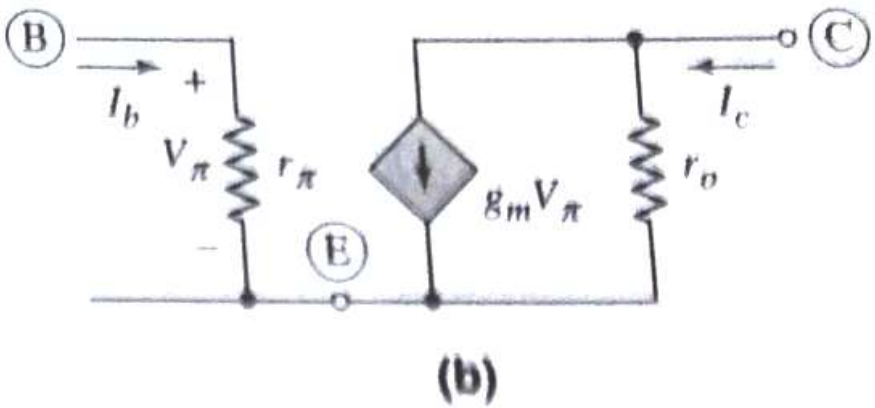




Small-Signal Hybrid-π model of pnp Transistor

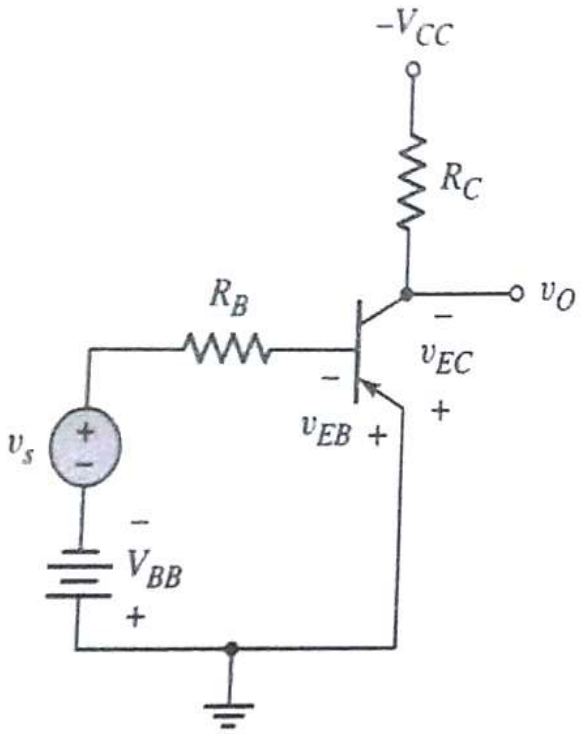


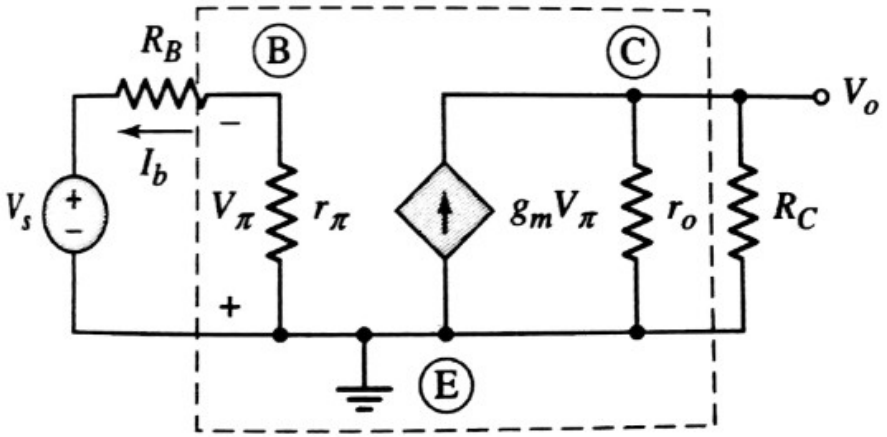
Original Circuit



Equivalent circuit with voltage polarities and current directions reversed

Small-Signal Hybrid-π model of pnp Transistor



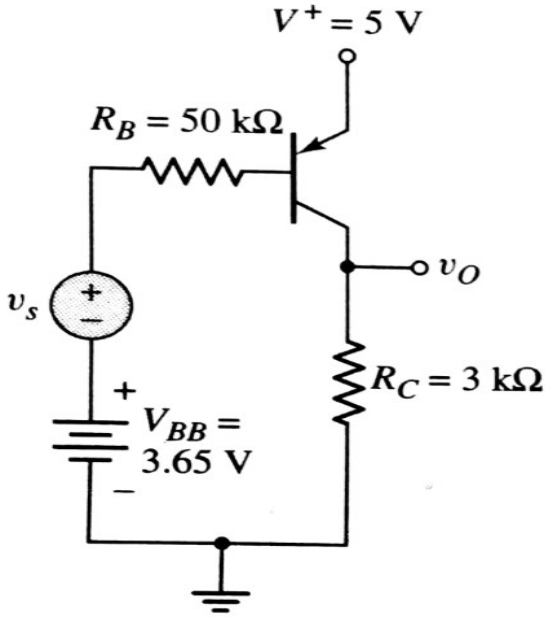


$$V_\pi = -\frac{V_s r_\pi}{R_B + r_\pi}$$

$$V_o = (g_m V_\pi)(r_o \| R_C)$$

$$Av = \frac{Vo}{Vs} = \frac{-g_m r_\pi}{r_\pi + RB}(Rc \,/\!/\, ro) = \frac{-\beta}{r_\pi + RB}(Rc \,/\!/\, ro)$$

same expression for pnp and npn transistor circuit



$\beta = 80,\ V_{EB(ON)} = 0.7V,\quad V_A = \infty$

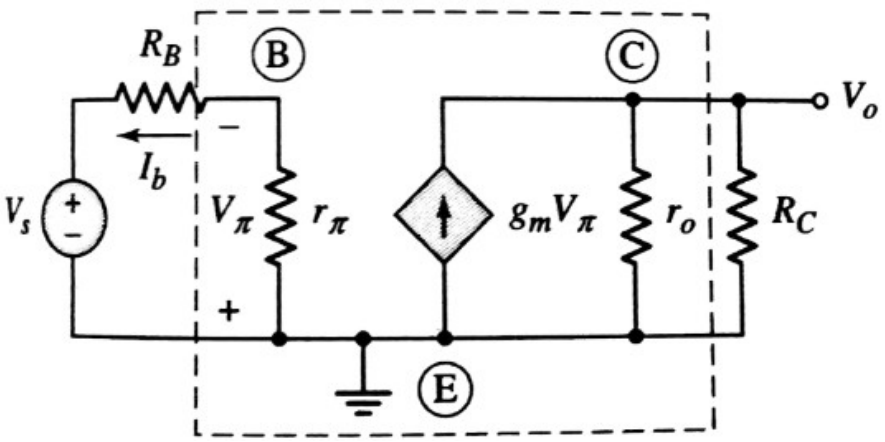
DC Analysis:

$V^+ - V_{EB} - I_{BQ} R_B - V_{BB} = 0$

$I_{BQ} = 13\ \mu A;\ I_{CQ} = 1.04mA;\ I_{EQ} = 1.05mA$

$V_{ECQ} = V^+ - I_{CQ} Rc = 1.88V$

AC Analysis: $g_m = \frac{I_{CQ}}{V_T} = \frac{1.04}{0.026} = 40\ mA/V$



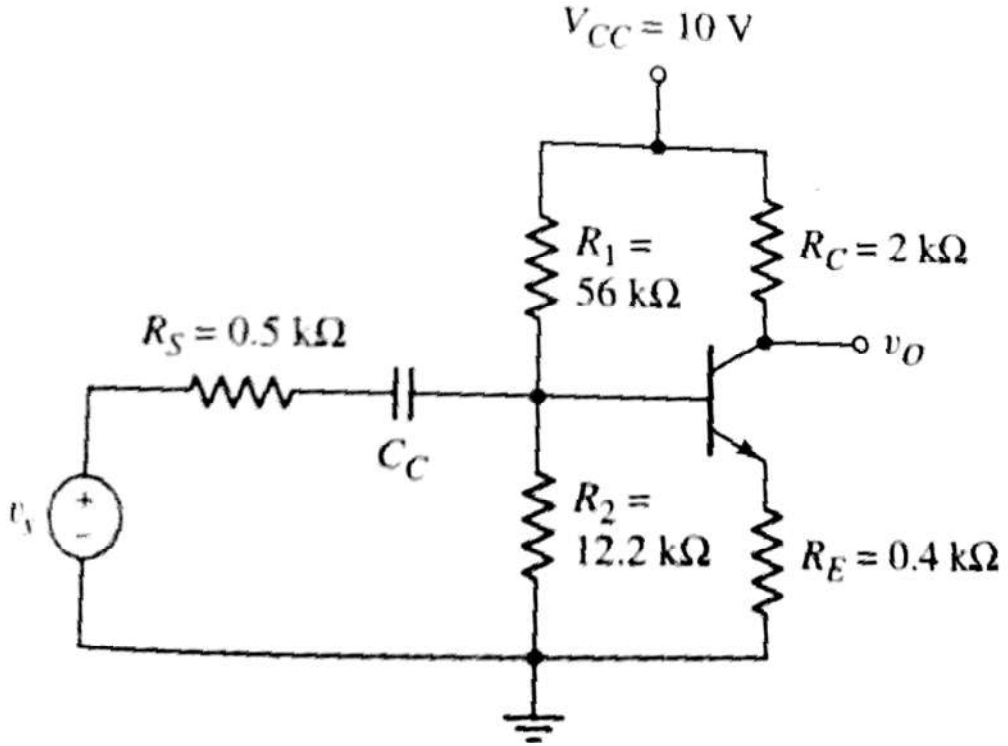
$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{80 * 0.026}{1.04} = 2\ K\Omega$

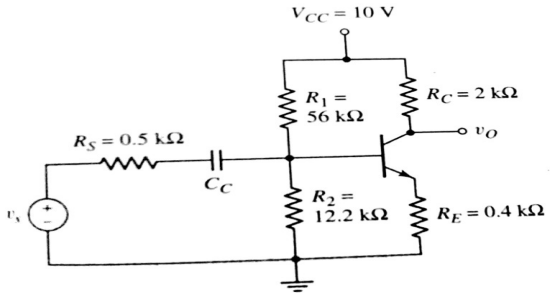
$r_O = \frac{V_A}{I_{CQ}} = \infty$

$Av = \frac{-\beta}{r_\pi + RB}(Rc) = -4.62$

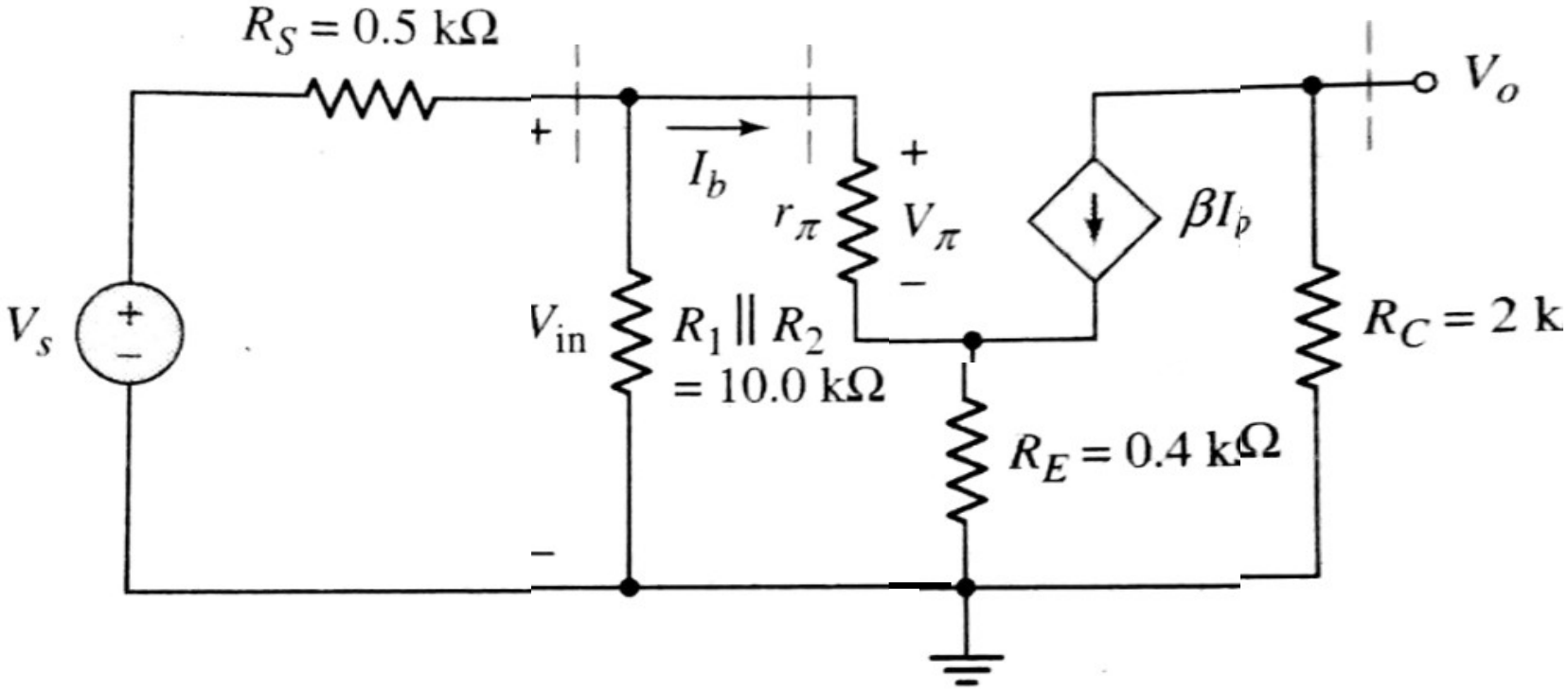
Small signal analysis of BJT Amplifier

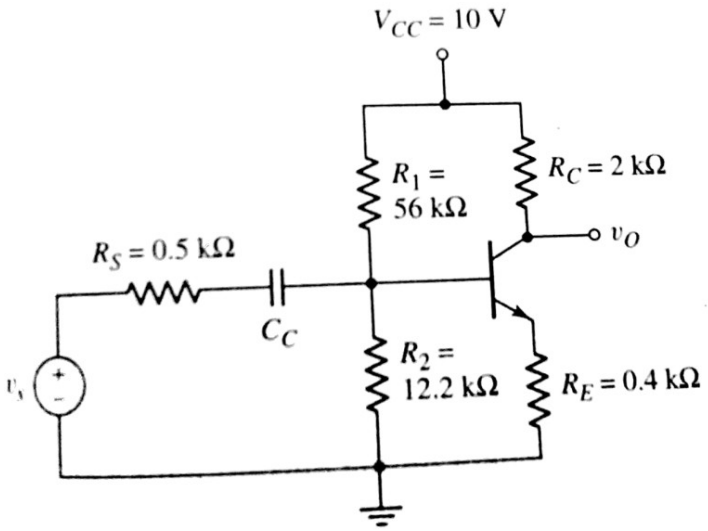
Solve Ex 6.1,Ex 6.2,Ex 6.3, TYU 6.1, TYU6.2



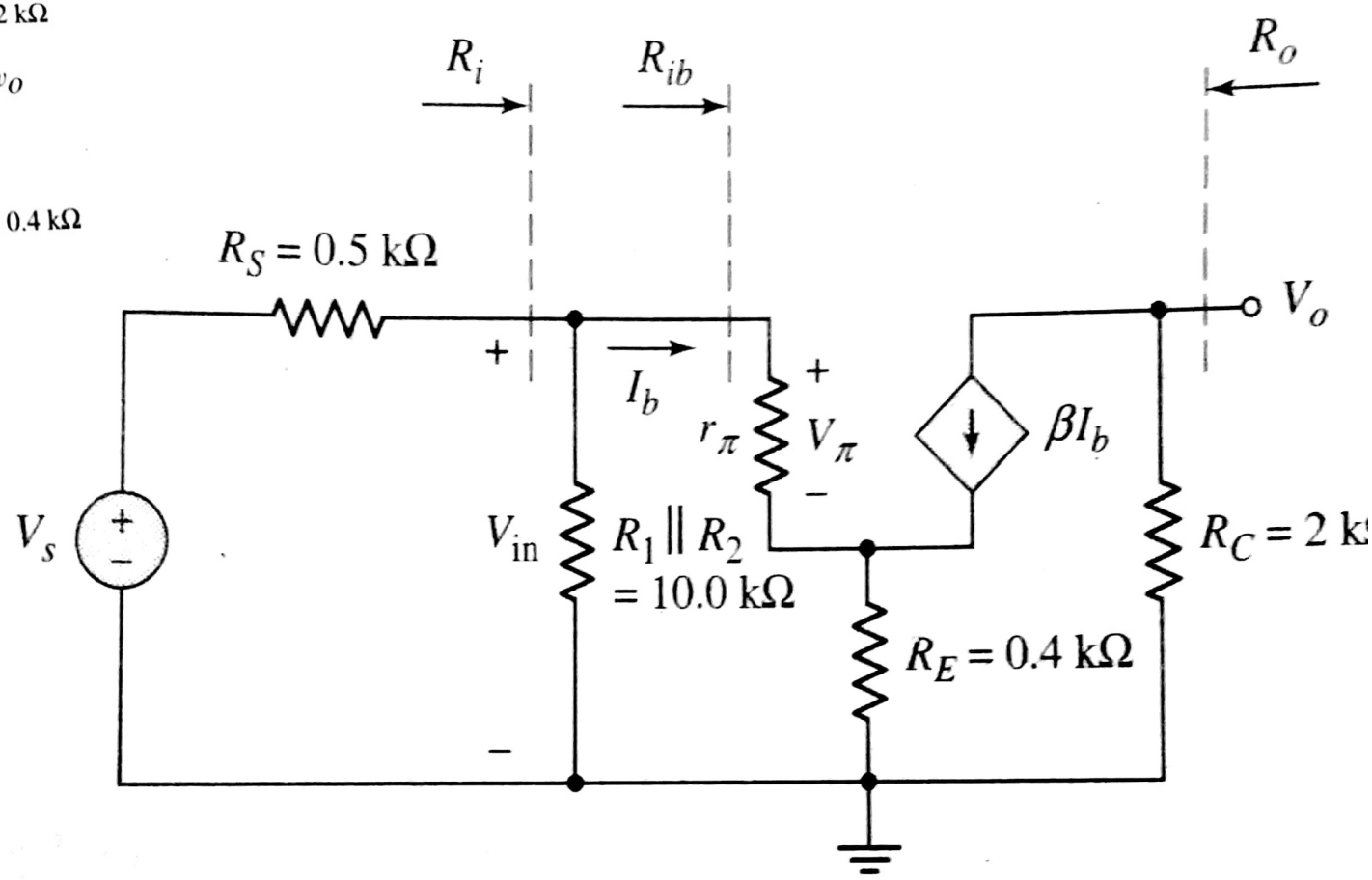


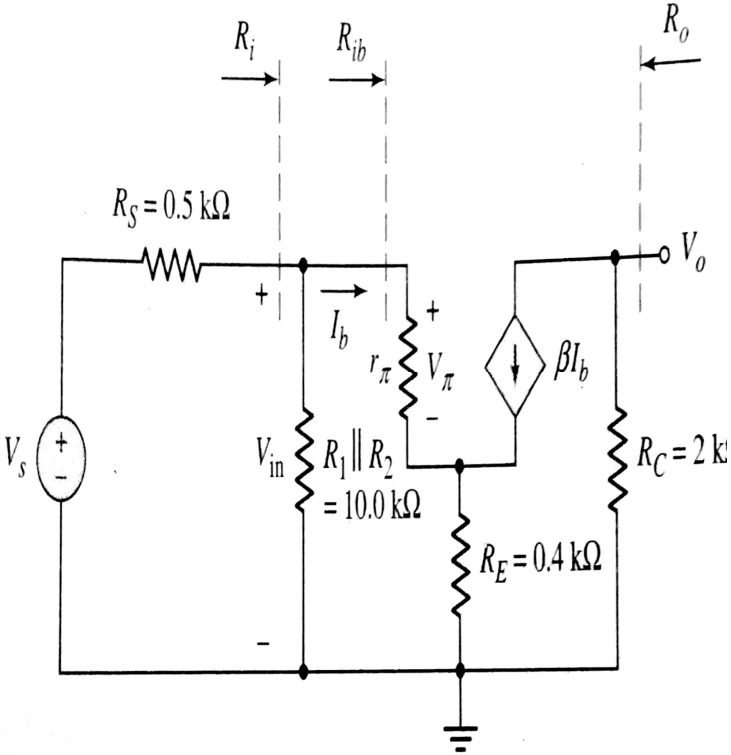
$\beta = 100$, $V_{BE(ON)} = 0.7V$, $V_A = \infty$





$\beta = 100$, $V_{BE(ON)} = 0.7V$, $V_A = \infty$

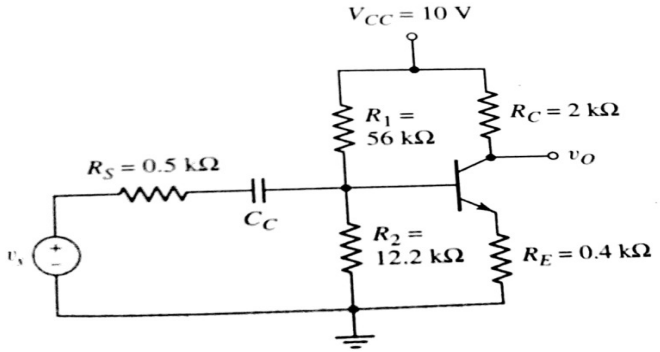


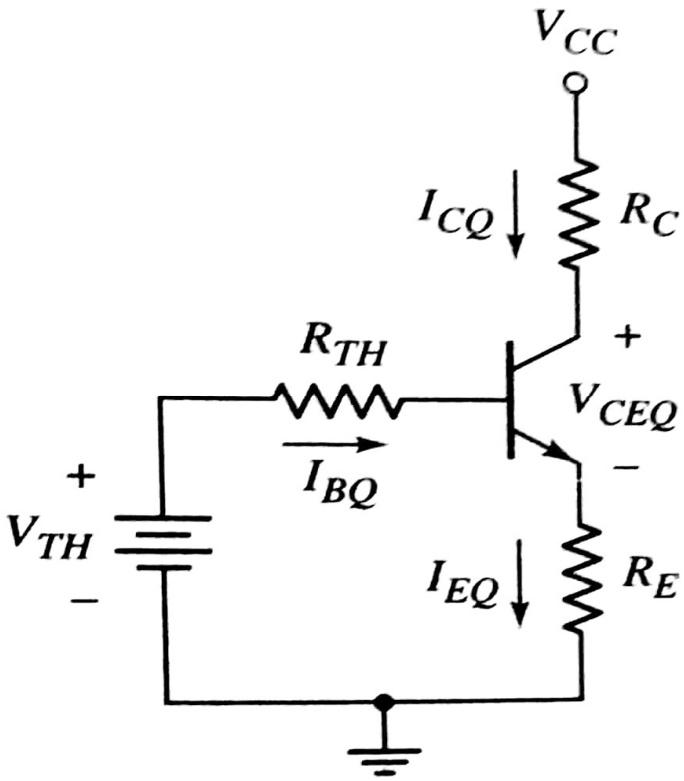


- $V_O = -(\beta Ib)R_C$
- $Vin = I_b r_\pi + (I_b + \beta Ib)\ R_E$
- $R_{ib} = \frac{V_{in}}{I_b} = r_\pi + (1+\beta)\ R_E$
- $R_i = R_1 \,//\, R_2 \,//\, R_{ib}$
- $V_{in} = \left(\frac{R_i}{R_i + Rs}\right) Vs$
- $Av = \frac{Vo}{V_s} = \frac{-(\beta I_b)RC}{V_s}$
- $= -(\beta R_C)\ \frac{V_{in}}{R_{ib}}\ \frac{1}{V_s}$
- $= \frac{-(\beta R_C)}{r_\pi + (1+\beta)\ R_E}\ \left(\frac{R_i}{R_i + R_S}\right)$
- If $R_i >> R_S$ and if $(1+\beta)\ R_E >> r_\pi$,

Then

$$Av = \frac{-(\beta R_C)}{(1+\beta)\ R_E} \cong \frac{-R}{R_E}$$





- DC Analysis:
- $R_{TH} = R_1 \,//\, R_2 = 10\ K$
- $$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2}$$
$$= 10\ (12.2/\ 68.2)\ = 1.79\ V$$

$$V_{TH} - I_{BQ} R_{TH} - V_{BE} - (1+\beta)\, I_{BQ} R_E = 0$$

$$I_{BQ} = 0.0216\ mA$$

$$I_{CQ} = 2.16\ mA$$

$$V_{CC} - I_{CQ} R_C - V_{CE} - (1+\beta)\, I_{BQ} R_E = 0$$

$$V_{CE} = 4.807 = 4.81\ V$$

- AC Analysis

$$g_m = \frac{I_{CQ}}{V_T} = \frac{2.16}{0.026} = 83.1 \text{ mA /V}$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{100 * 0.026}{2.16} = 1.2 \text{ K } \Omega \qquad r_O = \frac{V_A}{I_{CQ}} = \infty$$

$$R_{ib} = \frac{V_{in}}{I_b} = r_\pi + (1 + \beta) R_E$$

$$= 1.2 + (101)0.4 = 41.6 K\Omega$$

$$R_i = R_1 \,//\, R_2 \,//\, R_{ib}$$

$$= 10 \,//\, 41.6 = 8.06 \text{ K}\Omega$$

$$Av = \frac{Vo}{V_S} = \frac{-(\beta R_C)}{r_\pi + (1 + \beta) R_E} \left(\frac{R_i}{R_i + RS} \right)$$

$$= \frac{-(100 * 2)}{1.2 + (101)(0.4)} \left(\frac{8.06}{8.06 + 0.5} \right) = -4.53$$

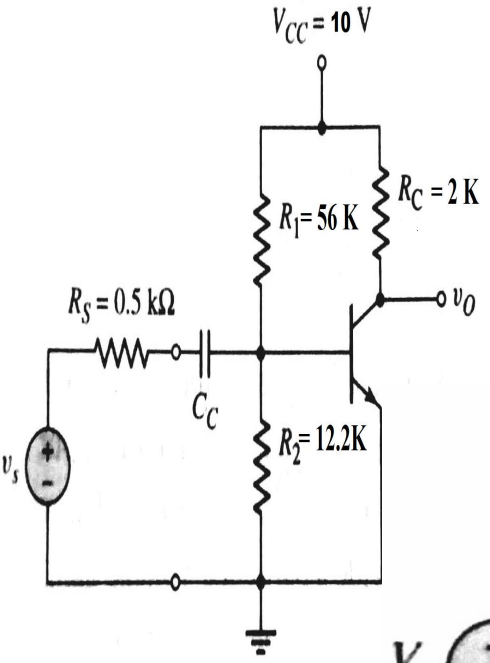
Bias Stability

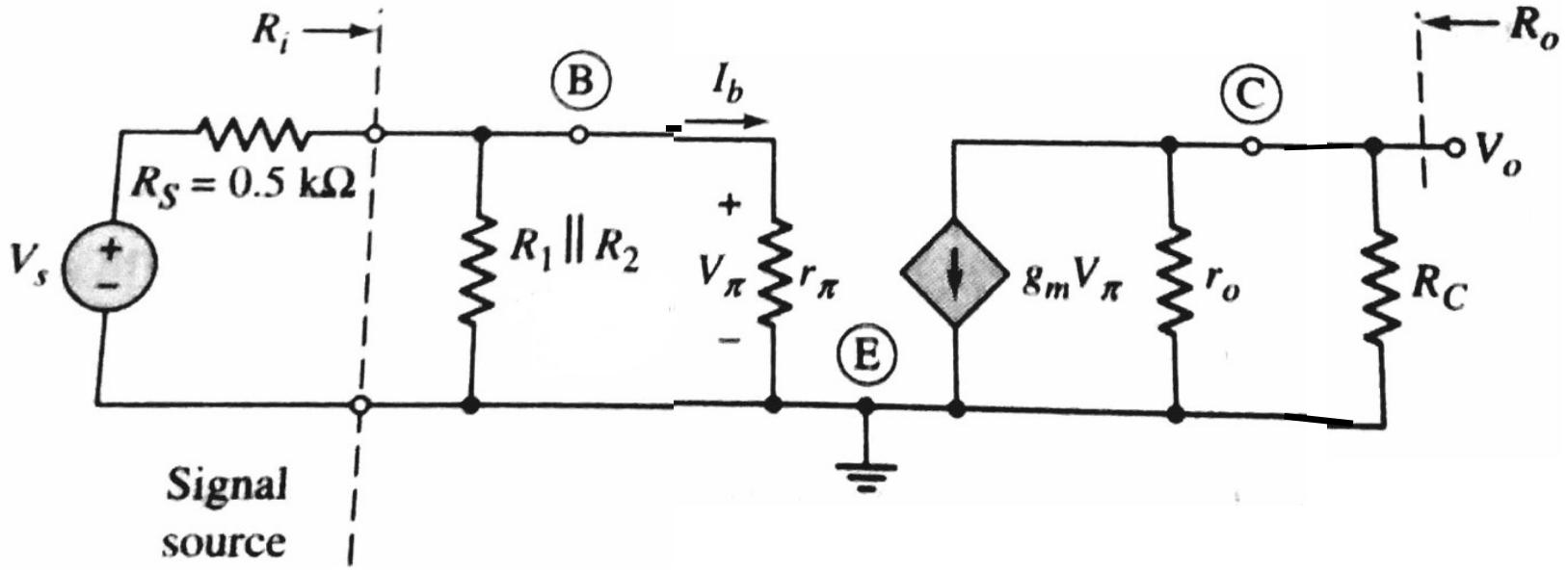
- Amplifier gain nearly independent of changes in current gain parameter β.

β	A_V
50	-4.41
100	-4.53
150	-4.57

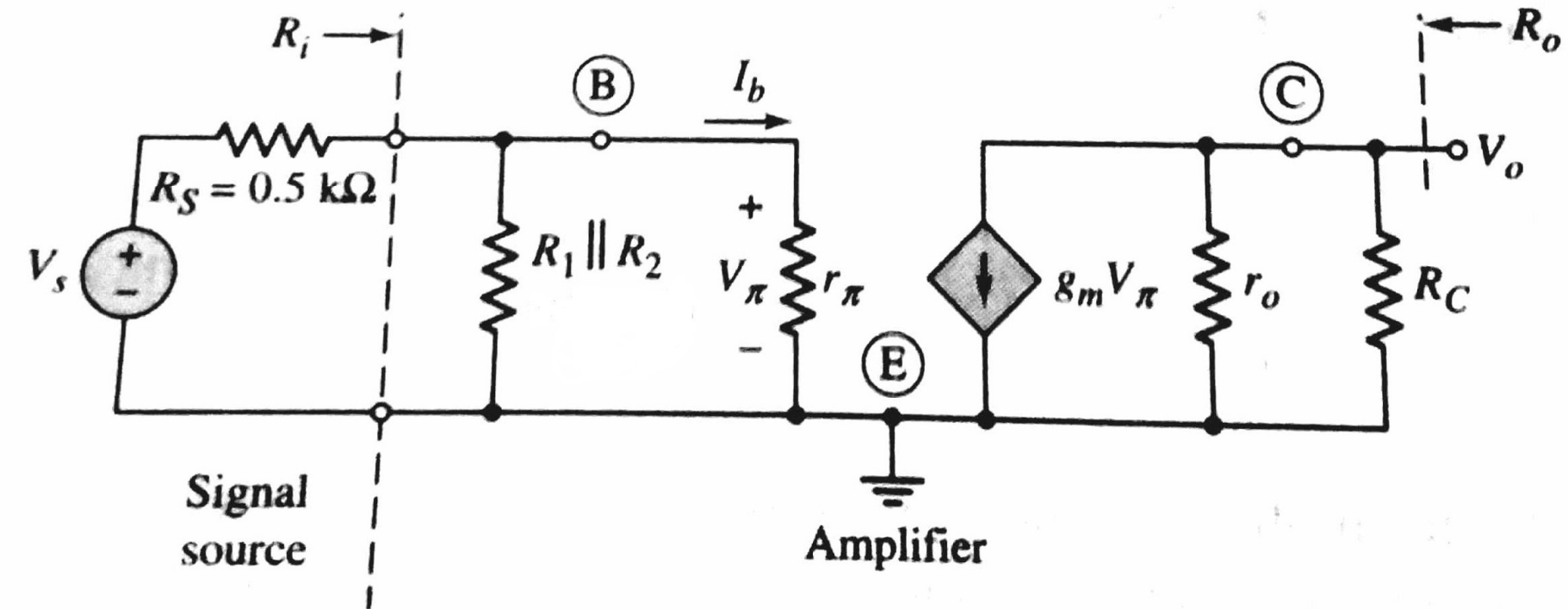
Common Emitter amplifier without R_E

$\beta = 100$, $V_{BE(ON)} = 0.7V$, $V_A = 100V$





Small signal equivalent circuit

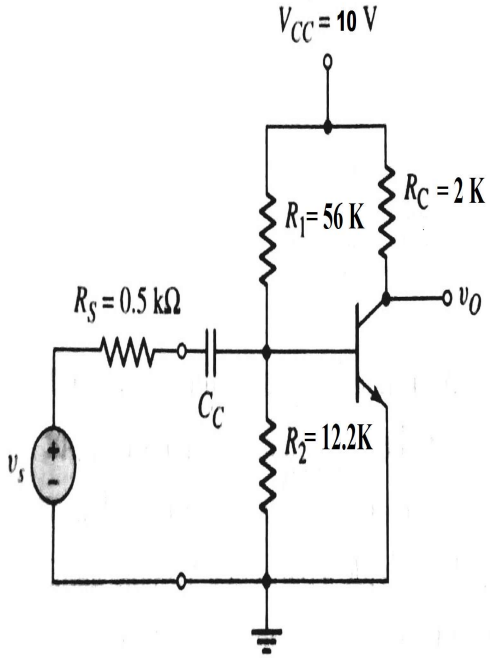


- DC Analysis: $R_{TH} = R_1 // R_2 = 10\ K$
- $V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2} = 10\ (12.2/\ 68.2) = 1.79\ V$
- $V_{TH} - I_{BQ} R_{TH} - V_{BE} = 0 \qquad I_{BQ} = 0.109\ mA \qquad I_{CQ} = 10.9\ mA$
- $V_{CC} - I_{CQ} R_C - V_{CE} = 0 \qquad V_{CE} = -11.8\ V$
- So transistor is not in active mode. Will not work as amplifier.
- AC Analysis

$$g_m = \frac{I_{CQ}}{V_T} = \frac{10.9}{0.026} = 419.23\ mA/V$$

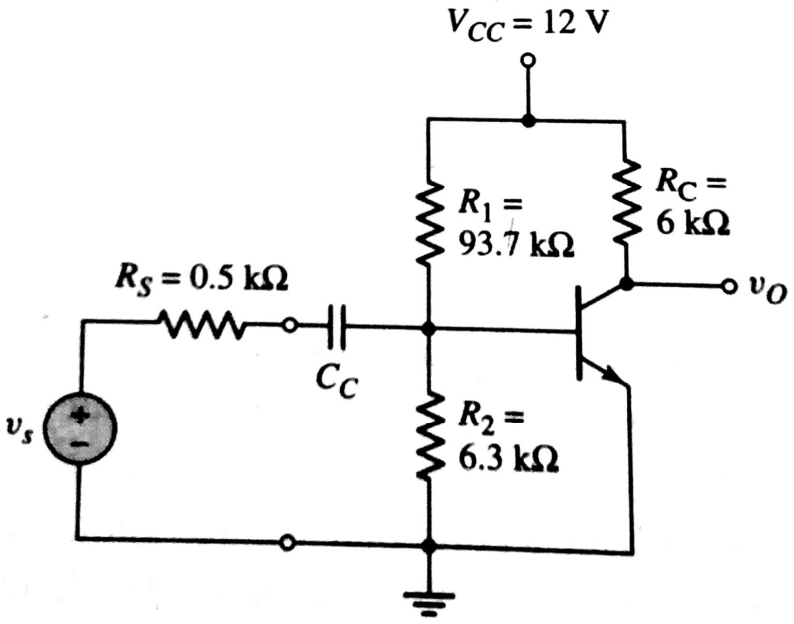
$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{100 * 0.026}{10.9} = 0.239\ K\ \Omega$$

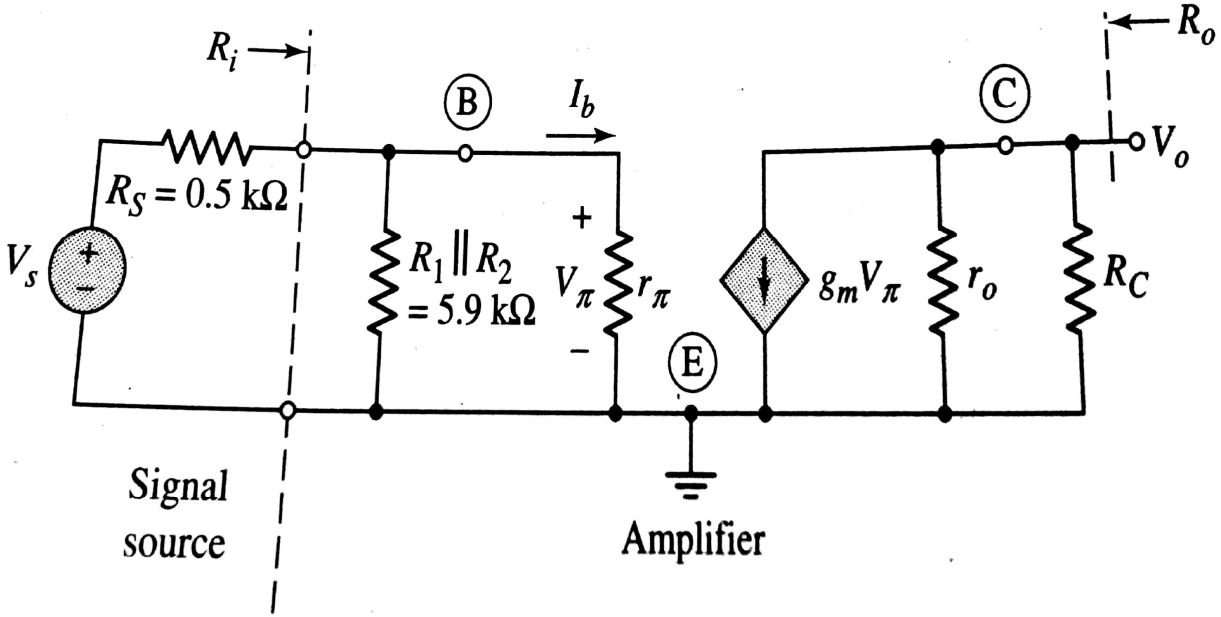
$$r_O = \frac{V_A}{I_{CQ}} = \frac{100}{10.9} = 9.17\ K\Omega$$



6.5 Common Emitter amplifier without R_E

$\beta = 100$, $V_{BE(ON)} = 0.7V$,
$V_A = 100V$





- DC Analysis: $R_{TH} = R_1 // R_2 = 5.9\ K$
- $V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2} = 10\ (12.2/\ 68.2) = 1.79\ V$
- $V_{TH} - I_{BQ} R_{TH} - V_{BE} = 0$ $I_{BQ} = 0.0095\ mA$ $I_{CQ} = 0.95\ mA$
- $V_{CC} - I_{CQ} R_C - V_{CE} = 0$ $V_{CE} = 6.3\ V$
- AC Analysis

$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.95}{0.026} = 36.5\ mA/V$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{100 * 0.026}{0.95} = 2.74\ K\Omega$$

$$r_O = \frac{V_A}{I_{CQ}} = \frac{100}{0.95} = 105\ K\Omega$$

$$Vo = -(g_m V_\pi)(r_o \,//\, R_C)$$

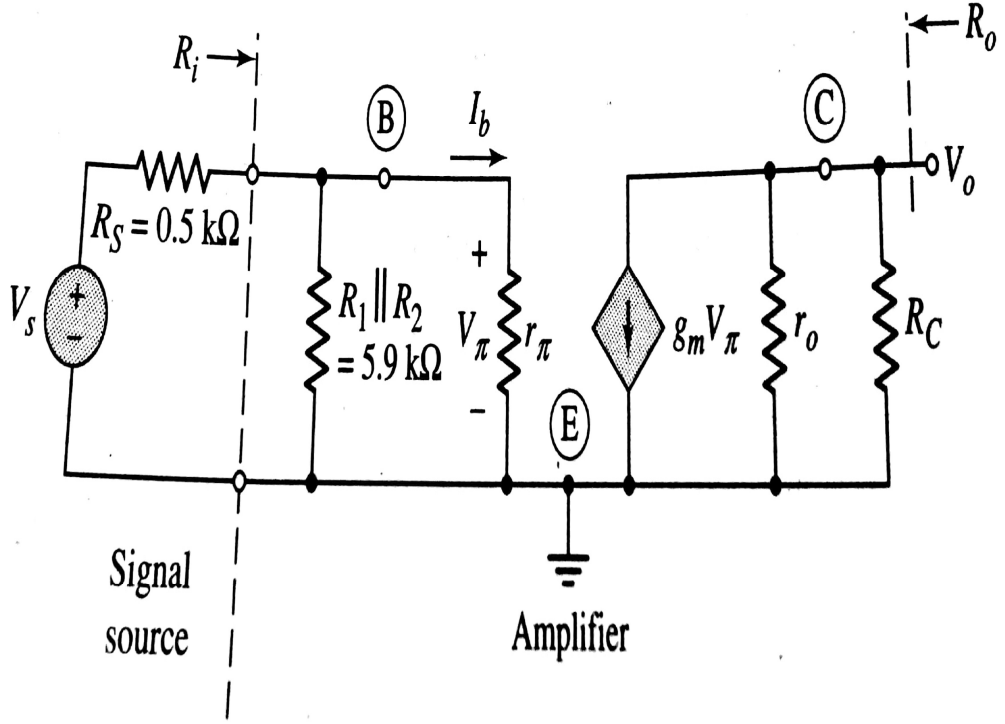
$$V\pi = Vs \left(\frac{R_1 \,//\, R_2 \,//\, R_\pi}{(R_1 \,//\, R_2 \,//\, R_\pi)+R_S}\right)$$

$$Av = \frac{Vo}{Vs} = -\, g_m\left(\frac{R_1 \,//\, R_2 \,//\, R_\pi}{(R_1 \,//\, R_2 \,//\, R_\pi)+R_S}\right)(r_o \,//\, R_C)$$

Av = -163

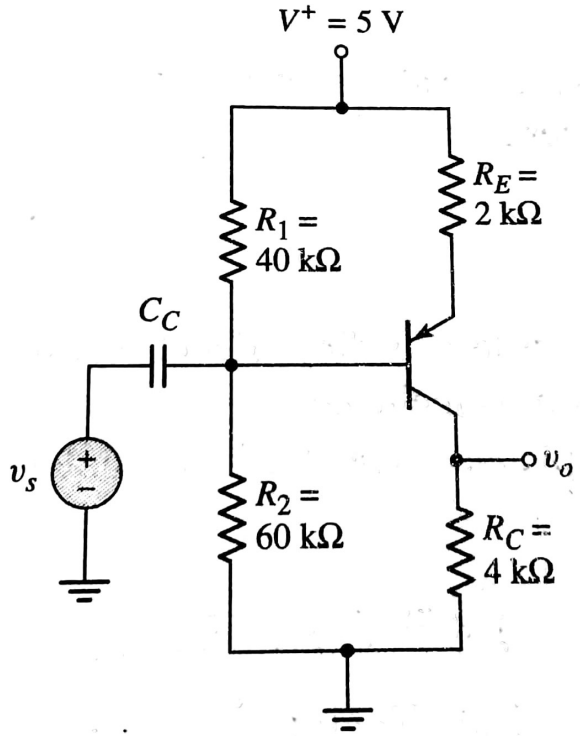
$Ri = R_1 \,//\, R_2 \,//\, R_\pi = 1.87\ K\Omega$

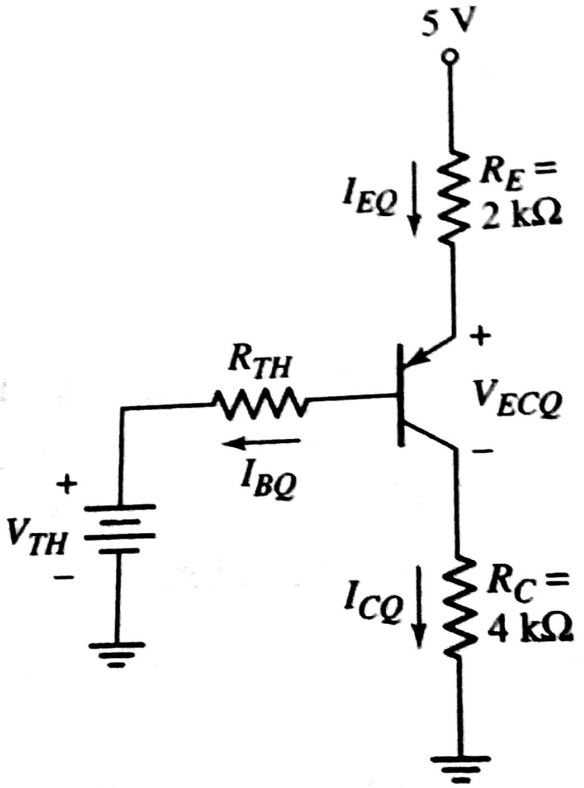
$R_O = (r_o \,//\, R_C) = 5.68\ K\Omega$



Ex 6.7 PNP CE amplifier with R_E

$\beta = 80$, $V_{EB(ON)} = 0.7V$, $V_A = \infty V$





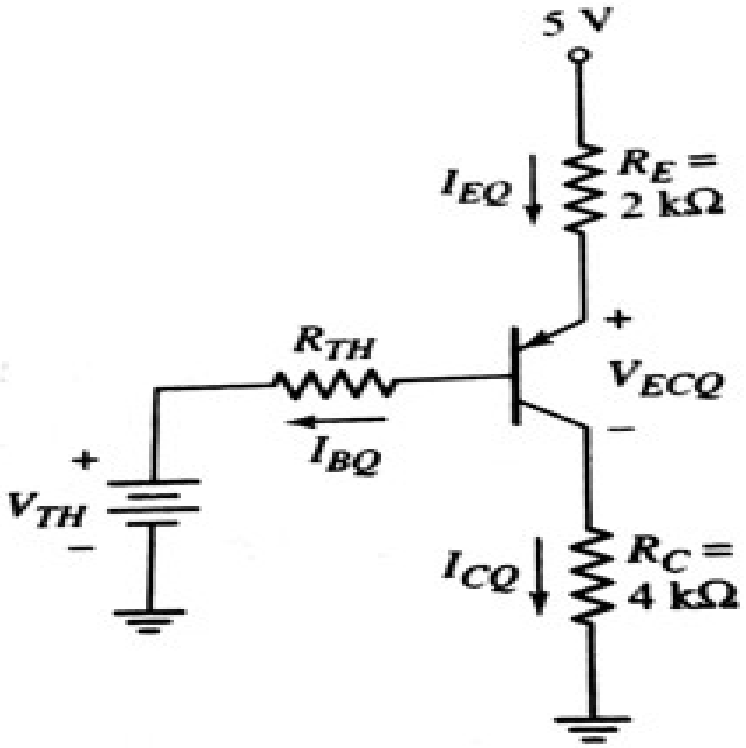
- $R_{TH} = R_1 \,//\, R_2 = 24\ K$
- $$V_{TH} = V_{CC}\frac{R_2}{R_1 + R_2} = 5\ (60/\ 100) = 3\ V$$
- $V^+ = V_{EB} + I_{BQ} R_{TH} + V_{TH} + (1+\beta) I_{BQ} R_E$

- I_{BQ} =6.99 μA; I_{CQ} = 0.559 mA; I_{EQ} =0.566 mA
- $V_{ECQ} = V^+ - I_{CQ} Rc - I_{EQ} R_E = 1.63\ V$
- AC Analysis

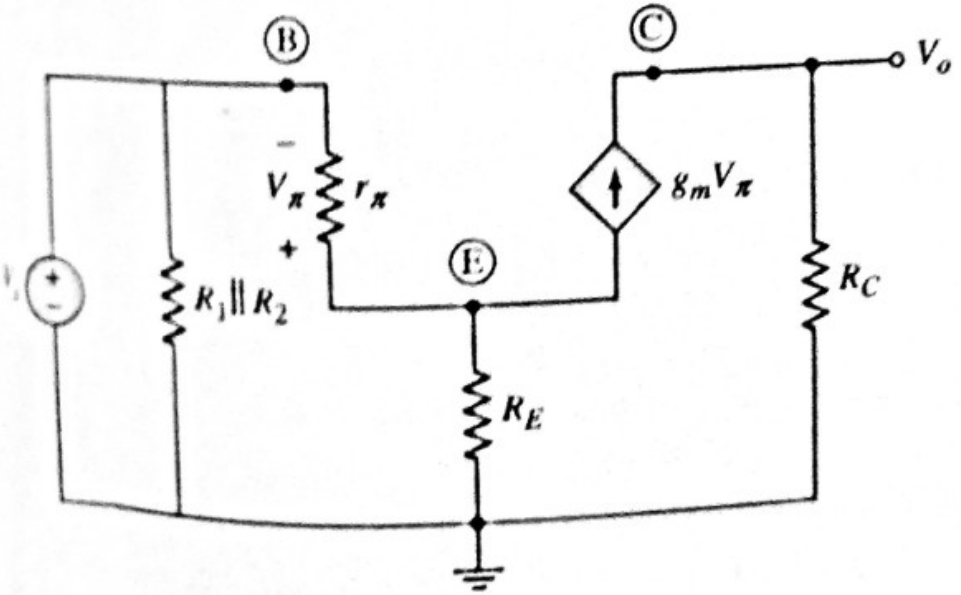
$$g_m = \frac{I_{CQ}}{V_T} = \frac{0.575}{0.026} = 22.115\ mA/V$$

$$r_\pi = \frac{\beta V_T}{I_{CQ}} = \frac{100 * 0.026}{0.575} = 4.52\ K\ \Omega$$

$$r_O = \frac{V_A}{I_{CQ}} = 100 / 0.575 = 173.91\ K$$



$$V_s = -V_\pi - \left(\frac{V_\pi}{r_\pi} + g_m V_\pi\right) R_E$$



- Vo = -(g_m V_π R_C)
- g_m r_π = β
- $V_\pi = \dfrac{-V_s}{1 + R_E(\frac{1+\beta}{r_\pi})}$
- A_V = -1.93

- Thumb rule $A_V = -\dfrac{RC}{R_E}$ = -2

Solve
Ex6.5, Ex 6.6, Ex 6.7
TYU 6.3, 6.4, 6.5, 6.6